Makrolon® T4065

/	MVR (300 °C/1.2 kg) 17 cm³/10 min; low viscosity; easy release; impact modified; enhanced chemical resistance; injection molding
ISO Shortname	PC-I

	Property	Test Condition	Unit	Standard	typical Value
	Rheological properties				
	Melt volume-flow rate	300 °C/ 1.2 kg	cm³/10 min	ISO 1133	17
	Molding shrinkage, parallel/normal	Value range based on general practical experience	%	b.o. ISO 2577	0.6 - 0.8
	Mechanical properties (23 °C/50 % r. h.)				
C	Tensile modulus	1 mm/min	MPa	ISO 527-1,-2	2330
	Yield stress	5 mm/min	MPa	ISO 527-1,-2	60
	Yield strain	5 mm/min	%	ISO 527-1,-2	6
C	Stress at break	5 mm/min	MPa	ISO 527-1,-2	57
C	Strain at break	5 mm/min	%	ISO 527-1,-2	90
	Flexural modulus	2 mm/min	MPa	ISO 178	2400
	Flexural strength	2 mm/min	MPa	ISO 178	94
	Charpy notched impact strength	23 °C	kJ/m²	ISO 7391/b.o. ISO 179/1eA	63P
	Charpy notched impact strength	-30 °C	kJ/m²	ISO 7391/b.o. ISO 179/1eA	33C
	Izod notched impact strength	23 °C	kJ/m²	ISO 7391/b.o. ISO 180/A	55P
	Izod notched impact strength	-30 °C	kJ/m²	ISO 7391/b.o. ISO 180/A	18C
C	Puncture impact properties - maximum force	23 °C	N	ISO 6603-2	4900
C	Puncture impact properties - maximum force	-30 °C	N	ISO 6603-2	6100
	Thermal properties				
C	Temperature of deflection under load	1.80 MPa	°C	ISO 75-1,-2	119
C	Temperature of deflection under load	0.45 MPa	°C	ISO 75-1,-2	133
C	Vicat softening temperature	50 N; 50 °C/h	°C	ISO 306	139
	Vicat softening temperature	50 N; 120 °C/h	°C	ISO 306	141
	Glow wire test (GWFI)	1.0 mm	°C	IEC 60695-2-12	850
	Glow wire test (GWFI)	1.5 mm	°C	IEC 60695-2-12	850
	Coefficient of linear thermal expansion, parallel	23 to 55 °C	10^{-4}/K	b.o. ISO 11359-1,-2	0.7
	Coefficient of linear thermal expansion, normal	23 to 55 °C	10^{-4}/K	b.o. ISO 11359-1,-2	0.7
	Other properties (23 °C)				
C	Density		kg/m³	ISO 1183-1	1200
	Processing conditions for test specimens				
C	Injection molding - Melt temperature		°C	ISO 294	290
C	Injection molding - Mold temperature		°C	ISO 294	80
C	Injection molding - Injection velocity		mm/s	ISO 294	200

C These property characteristics are taken from the CAMPUS plastics data bank and are based on the international catalogue of basic data for plastics according to ISO 10350.

Impact properties: N = non-break, P = partial break, C = complete break

Disclaimer

Information Impact properties

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Typical value

These values are typical values only. Unless explicitly agreed in written form, the do not constitute a binding material specification or warranted values. Values may be affected by the design of the mold/die, the processing conditions and coloring/pigmentation of the product. Unless specified to the contrary, the property values given have been established on standardized test specimens at room temperature.

General

The manner in which you use and the purpose to which you put and utilize our products, technical assistance and information (whether verbal, written or by way of production evaluations), including any suggested formulations and recommendations are beyond our control. Therefore, it is imperative that you test our products, technical assistance, information and recommendations to determine to your own satisfaction whether our products, technical assistance and information are suitable for your intended uses and applications. This application-specific analysis must at least include testing to determine suitability from a technical as well as health, safety, and environmental standpoint. Such testing has not necessarily been done by Covestro. Unless we otherwise agree in writing, all products are sold strictly pursuant to the terms of our standard conditions of sale which are available upon request. All information and technical assistance is given without warranty or guarantee and is subject to change without notice. It is expressly understood and agreed that you assume and hereby expressly release us from all liability, in tort, contract or otherwise, incurred in connection with the use of our products, technical assistance, and information. Any statement or recommendation not contained herein is unauthorized and shall not bind us. Nothing herein shall be construed as a recommendation to use any product in conflict with any claim of any patent relative to any material or its use. No license is implied or in fact granted under the claims of any patent. With respect to health, safety and environment precautions, the relevant Material Safety Data Sheets (MSDS) and product labels must be observed prior to working with our products.

Non Medical and non Food Contact Grade

This product is not designated for the manufacture of a pharmaceutical/medicinal product, medical device or of intermediate products for medical devices1). This product is also not registered for Covestro for the use in other specifically regulated applications, in particular applications requiring regulatory registration, approval or notification (e.g. including cosmetics, plant protection, food processing, food contact and others). If the intended use of the product is for the manufacture of a pharmaceutical, medical device or of intermediate products for medical devices or for other specifically regulated applications which may lead to a regulatory obligation of Covestro, Covestro must be contacted in advance to provide its agreement to sell such product for such purpose. Nonetheless, any determination as to whether a product is appropriate for use in a pharmaceutical, medical device or intermediate products for medical devices or for the use in other specifically regulated applications, must be made solely by the purchaser of the product without relying upon any representations by Covestro, irrespective of the existence of any regulatory obligation for the registration, approval or notification. 1) Please see the "Guidance on Use of Covestro Products in a Medical Application" document.

Recommended Processing and Drying Conditions

Barrel temperatures are valid for a standard 3-zone barrel. Temperature set-up for different barrel types may change according to configuration. Values for hold pressure as percentage of injection pressure may vary depending on, amongst others, part geometry, injection molding machine and injection mold. Drying conditions are for dry air dryers only. Drying times and drying temperatures may differ depending on valid dryer type. Further information is provided by your local Covestro support as well as in the following brochures: Injection Molding of High Quality Molded Parts - Drying; Determining the Dryness of Makrolon by TVI Test; The fundamentals of shrinkage in thermoplastics; Shrinkage and deformation of glass fiber reinforced thermoplastics [...]. https://www.plastics.covestro.com/Library/Overview.aspx

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